

Lecture 5 – Counting Theories & Naive Bayes

INT104 Artificial Intelligence

Counting Theories & Naive Bayes

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Start with rolling a dice

- When we roll a dice for 60 times, the results we expected are:

Case	1	2	3	4	5	6
times	10	10	10	10	10	10

which will almost never happen

- What we get is usually

Case	1	2	3	4	5	6
times	12	7	11	8	9	13



Does that mean the knowledge we have $p(X = i) = \frac{1}{6}$ is wrong?

Bayesian

- Not necessary
- The knowledge we have (or say the results we expect) is considered as prior probability (or prior knowledge)
- The results we obtained in the experiment are considered as posterior probability



Unlike Frequentist, Bayesian considers the distribution of unmet data besides observed experiments

Bayes' Rule

- The famous Bayes' Rule states the relationship between prior probability distribution and posterior probability distribution

$$P(c|\mathbf{x}) = \frac{P(c)P(\mathbf{x} | c)}{P(\mathbf{x})} \quad (1)$$

where c is considered as a class, x is considered as a set of samples with engaged attributes

- $P(c)$ is named as prior probability
- $P(\mathbf{x}|c)$ is named as likelihood
- $P(c|\mathbf{x})$ is named as posterior probability
- $P(\mathbf{x})$ is considered as evidence factor (observation)
- Bayes' Rule can be used for various purposes such as parameter estimation, classification and model selection.

Model Likelihood

- Model likelihood represents how likely a model will be observed according to the given posterior and prior probability hence can be used for model selection.

$$\text{Likelihood} = \frac{\text{Posterior Probability} \times \text{Observation}}{\text{Prior Probability}} \quad (2)$$

- Given a dataset, the observation remain the same for different classes / samples.

Model Likelihood

- Rolling a dice is boring, we now roll a coin instead.
- Using Θ to represent the parameters in the model, calculate model likelihood for the following two models given the experiment results "THHTTTHTTH"
 - $p(\text{head}) = 0.5$ & $p(\text{tail}) = 0.5$
 - $p(\text{head}) = 0.4$ & $p(\text{tail}) = 0.6$

- Frequentists choose the model with highest likelihood as the best model, which is known as Maximum Likelihood Estimation (MLE).

Log-likelihood

- When the number of samples increases, even computer may not be able to calculate correctly.
- The Model likelihood of dataset D for a candidate model whose parameter sets are Θ has a likelihood of

$$\mathcal{L} = p(D|\Theta) = \prod_{x \in D} p(x|\Theta) \quad (3)$$

- For easier calculation, log-likelihood is commonly used which replace multiplications with additions i.e.

$$\mathcal{LL} = \log p(D|\Theta) = \sum_{x \in D} \log p(x|\Theta) \quad (4)$$

- The base of logarithm does not matter but a base 2 or 16 could lead to a stronger practical meaning in computer-based systems

Another Example

- For a set $D = \{16, 8, 2, 64\}$, which of the following rule has a higher likelihood?
 - h_{two} : This is a set containing powers of two under 100.
 - h_{even} : This is a set containing even numbers below 100.
- Occam's razor (size principle): the model favours the simplest (smallest) hypothesis consistent with the data

An example of showing overfit

- For a set $D = \{16, 8, 2, 64\}$, can you calculate the model likelihood for "The power of 2 under 100 except 32"?

- If a new sample collected is 32, can you calculate the posterior probability and model likelihood?

Likelihood & Entropy

- Suppose we have a dataset Q (as the training dataset), where a model Q proposed (in the form of the probability mass distribution of symbols) are $\mathbf{Q} = \{q_1, q_2, \dots, q_l\}$.
- Now there is a separate dataset P such that $\mathbf{P} = \{p_1, p_2, \dots, p_l\}$.
- Assume there are N samples in dataset P and M samples in dataset Q . The likelihood of dataset P can be written as

$$\prod_{i=1}^l p_i^{Np_i} \quad (5)$$

Likelihood & Entropy

- The likelihood can then be written in a logarithm scale then averaged as per sample

$$\mathcal{L} = \frac{1}{N} \log \prod_{i=1}^l p_i^{Np_i} = \frac{1}{N} \sum_{i=1}^l Np_i \log p_i = \sum_{i=1}^l p_i \log p_i \quad (6)$$

which is the (negative) entropy of P .

- Similarly, the likelihood of dataset P given model Q can be written as (in a scale of logarithm and on per sample basis)

$$\mathcal{L}' = \frac{1}{N} \log \prod_{i=1}^l q_i^{Np_i} = \frac{1}{N} \sum_{i=1}^l Np_i \log q_i = \sum_{i=1}^l p_i \log q_i \quad (7)$$

which is the (negative) entropy between P and Q .

Likelihood & Entropy

- Hence the difference between the cross entropy of P & Q and the entropy of P is

$$KL_{Div} = - \sum_{i=1}^l p_i \log q_i + \sum_{i=1}^l p_i \log p_i = - \sum_{i=1}^l p_i \log \frac{q_i}{p_i} \quad (8)$$

which is the relative entropy between P and Q . Relative entropy is also known as the Kullback–Leibler divergence.



KL divergence measures how much uncertainty is increased in the unobserved dataset.

Bayesian Estimation & Laplacian Correction

- Considering the fact that if a case never appeared in the training dataset, how the model likelihood will be effected?
- This is known as a zero–point problem
- A smooth process can be performed by introducing prior probability distribution, which is known as Bayesian Estimation
- More commonly, an uniform distribution is used as the prior probability distribution, in which case, it is called Laplacian Correction

Bayes' Rule for Classification

- How can we make use of Bayes' Rule for Classification?
 - We want to maximise the posterior probability of observations
 - This method is named MAP estimation (Maximum a posteriori)
- The posterior is simply the likelihood times the prior and then normalised, i.e.

$$p(c|\mathbf{x}) = \frac{p(c)p(\mathbf{x} | c)}{p(\mathbf{x})} = \frac{p(c)}{p(\mathbf{x})} \prod_{i=1}^l p(x_i | c) \quad (9)$$

where $\mathbf{x} = (x_1, x_2, \dots, x_l)$ are considered as a sample / samples with engaged attributes

- As the observation is same (the same training dataset), we have

$$p(c|\mathbf{x}) \propto p(c)p(\mathbf{x}|c) \quad (10)$$

Bayes' Rule for Classification

- So we need to find the likelihood and the prior probability

- According to Law of Large Numbers, the prior probability can be taken as the probability resulted from the frequency of observations
 - So we only care about how to find the model likelihood
 - As we have no candidate models yet, it is not easy to find model likelihood as the probability of observing samples according to an existing model
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Naive Bayes Classifier

- As the class distributions with different conditions are observable, we can find $p(\mathbf{x}|c)$ directly
- Naive Bayes Classifier simplifies the way to calculate $p(\mathbf{x}|c)$ by assuming elements in the feature vector (attributes) $\mathbf{x}$ are independent to each other, i.e.

$$p(\mathbf{x}|c) = p(x_1|c)p(x_2|c) \dots p(x_l|c) \quad (11)$$

where $\mathbf{x} = (x_1, x_2, \dots, x_l)$

Example

- We have the following information

Outlook	Temperature	Humidity	Windy	Play
Overcast	Hot	High	False	Yes
Overcast	Cool	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Rainy	Mild	High	False	Yes
Rainy	Cool	Normal	False	Yes
Rainy	Cool	Normal	True	No
Rainy	Mild	Normal	False	Yes
Rainy	Mild	High	True	No
Sunny	Hot	High	True	No
Sunny	Hot	High	False	No
Sunny	Mild	High	False	No
Sunny	Cool	Normal	False	Yes
Sunny	Mild	Normal	True	Yes

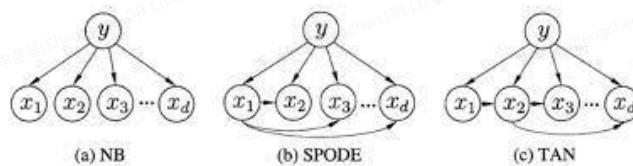
- Will you play on the day of Mild?

Example

- Will the following condition be considered as a proper day for playing?
 - Sunny, Windy
 - Overcast, Normal Humidity & Cool
- What about the following case?
 - Overcast, Windy & Cool

Semi-Naive Bayes Classifier

- Naive Bayes assumes all conditions / features are independent to each other, which never happens
- Semi-Naive Bayes allows dependencies between features



Bayesian Network

The dependencies between features could be very complicated.

- Bayesian Network uses Directed Acyclic Graph (DAG) to show such dependencies, which is also known as Bayesian Graphical Model
- The idea of Bayesian Graphical model is to factorise the joint probability distribution to more readable conditional probability distribution, which benefits the modelling algorithm (especially for semantic modelling)
- The learning of structure, parameters and queries are all academic problems